Mechanical Chaos

A shaken pendulum that can’t make up its mind

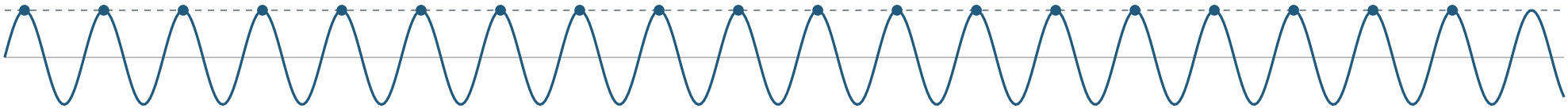
One pendulum, settings untouched, 68 minutes: **wild**, **calm**, **wild again**, **calm**. **What changed?**



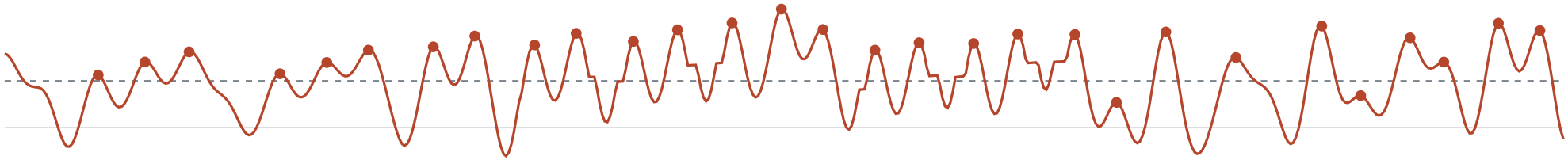
Parth Bhargava · PHY 474 Advanced Laboratory I · October 2026

A pendulum should repeat itself. Ours does not.

A clock pendulum: every peak reaches the same height (dashed)



Our pendulum, shaken, on the same scale: peaks from **half** to **two and a half times** that height



Chaos: fixed rules, yet tiny differences in the start grow until the future can't be predicted

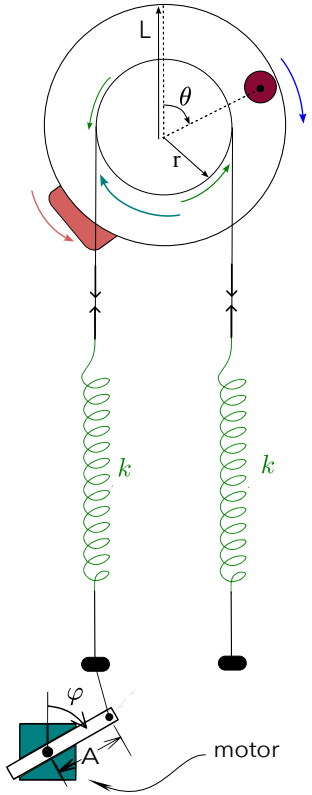
Question: how much more repeatable does the motion become, and does a test support chaos?

Short answer: 4.6 times more repeatable; consistent with transient chaos, not proof of it

Adding up the torques gives the Duffing equation

Torques on the disk[a]:

gravity on mass	$+mgL\sin\theta$
magnet	$-b\dot{\theta}$
two springs	$-2kr^2\theta$
motor arm	$+krA\cos(\Omega t)$



$$I\ddot{\theta} = mgL\sin\theta - b\dot{\theta} - 2kr^2\theta + krA\cos(\Omega t)$$

$$\implies I\ddot{\theta} + b\dot{\theta} + 2kr^2\theta - mgL\sin\theta = F_0\cos(\Omega t)$$

$$\text{Let } F_0 \equiv krA \text{ and } \sin\theta \approx \theta - \tfrac{\theta^3}{6}$$

$$\implies I\ddot{\theta} + b\dot{\theta} + (2kr^2 - mgL)\theta + \tfrac{mgL}{6}\theta^3 = F_0\cos(\Omega t)$$

which is equivalent to

$$\ddot{\theta} + \delta\dot{\theta} + \alpha\theta + \beta\theta^3 = \gamma\cos(\Omega t)$$

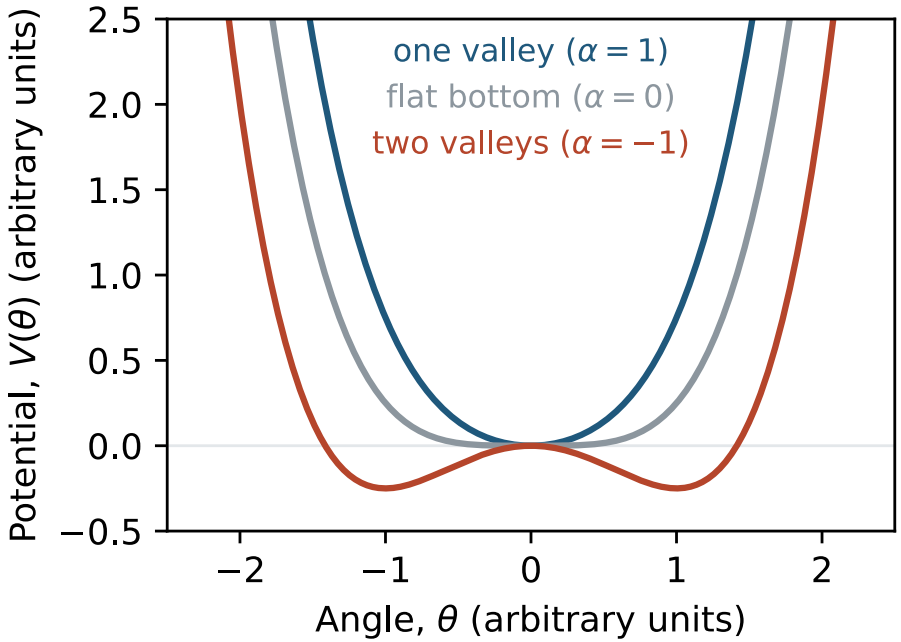
Forced Duffing's Equation

[a] θ is measured from the mass-up position and I is the moment of inertia of disk and mass. Dividing by I gives the Duffing coefficients. The cubic form holds up to about 1 rad; the disk swings further, so the results read the potential from data instead.

Each term of the equation is a part of our pendulum

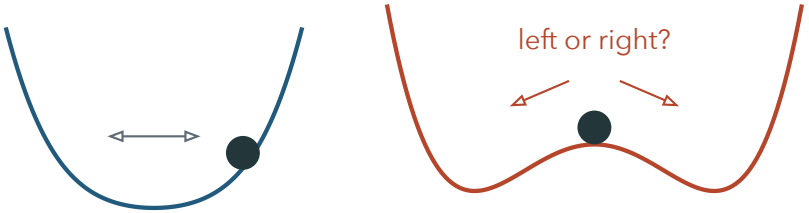
$$\underbrace{\ddot{\theta}}_{\text{acceleration}} + \underbrace{\delta\dot{\theta}}_{\text{friction}} + \underbrace{\alpha\theta + \beta\theta^3}_{\text{potential}} = \underbrace{\gamma\cos(\Omega t)}_{\text{shaking}}$$

Term	Coefficient	Set by
friction	$\delta = \frac{b}{I}$	magnet: gap h
potential	$\alpha = \frac{2kr^2 - mgL}{I}$, $\beta = \frac{mgL}{6I}$	springs against gravity on the mass m
shaking	$\gamma = \frac{krA}{I}$, Ω	motor: arm A, voltage V



The potential, $V(\theta) = \frac{1}{2}\alpha\theta^2 + \frac{1}{4}\beta\theta^4$

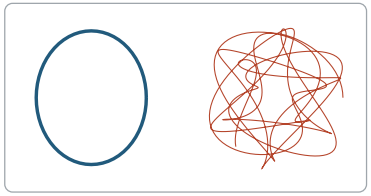
Chaos needs the drive's phase, and has to be measured



one valley
ball rocks back and forth

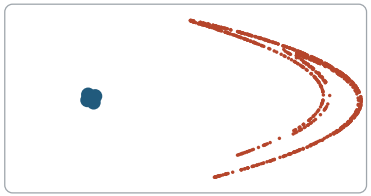
two valleys
a tiny nudge decides the side

Two routes: hop between two valleys[1], or shake one valley hard[2]. The drive's phase is a third variable: the same angle and speed can evolve differently. Either route can be **temporary**[3].



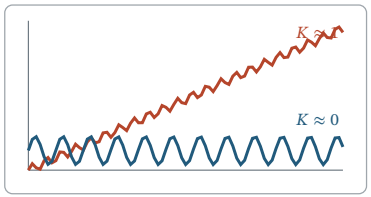
Signature 1 · never repeats

periodic: one closed loop; irregular: never closes



Signature 2 · strobe width

flash once per drive cycle: the cloud's width measures repeatability[4]

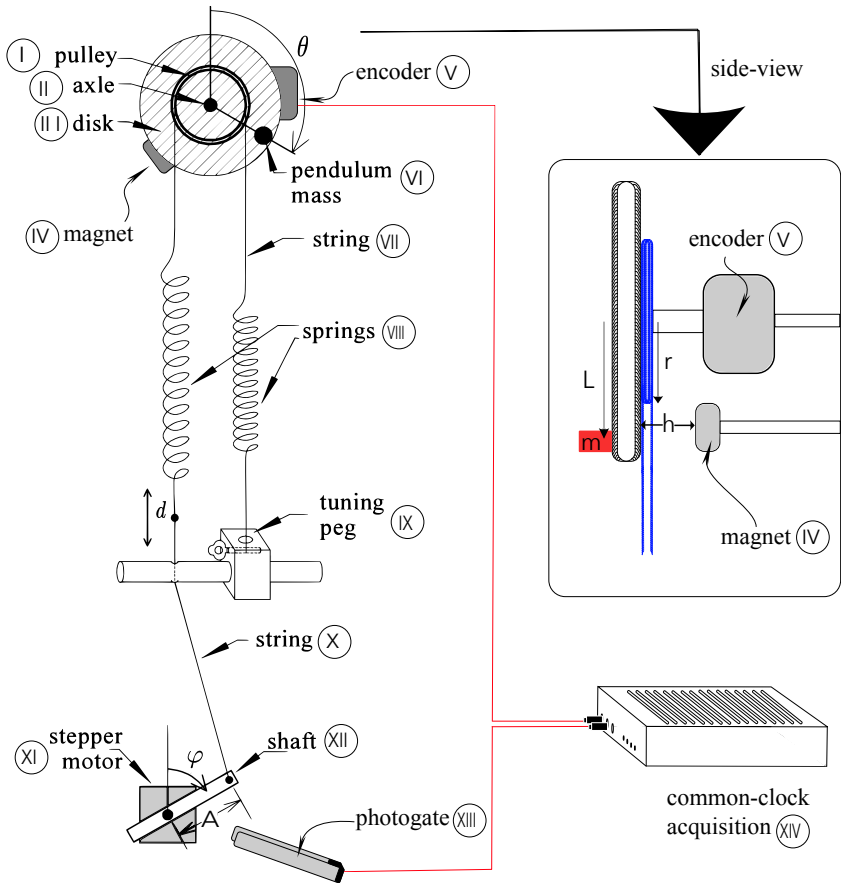


Signature 3 · 0–1 test

a weighted sum stays bounded, $K \approx 0$, or grows, $K \approx 1$[5]. Noise also gives 1

[1] F. C. Moon and P. J. Holmes, A magnetoelastic strange attractor, Journal of Sound and Vibration 65, 275 (1979)
[2] Y. Ueda, Randomly transitional phenomena in the system governed by Duffing's equation, Journal of Statistical Physics 20, 181 (1979)
[3] A. S. de Paula, M. A. Savi, and F. H. I. Pereira-Pinto, Chaos and transient chaos in an experimental nonlinear pendulum, Journal of Sound and Vibration 294, 585 (2006)
[4] R. DeSerio, Chaotic pendulum: The complete attractor, American Journal of Physics 71, 250 (2003)
[5] G. A. Gottwald and I. Melbourne, On the Implementation of the 0-1 Test for Chaos, SIAM Journal on Applied Dynamical Systems 8, 129 (2009)

The real setup: four knobs and two sensors

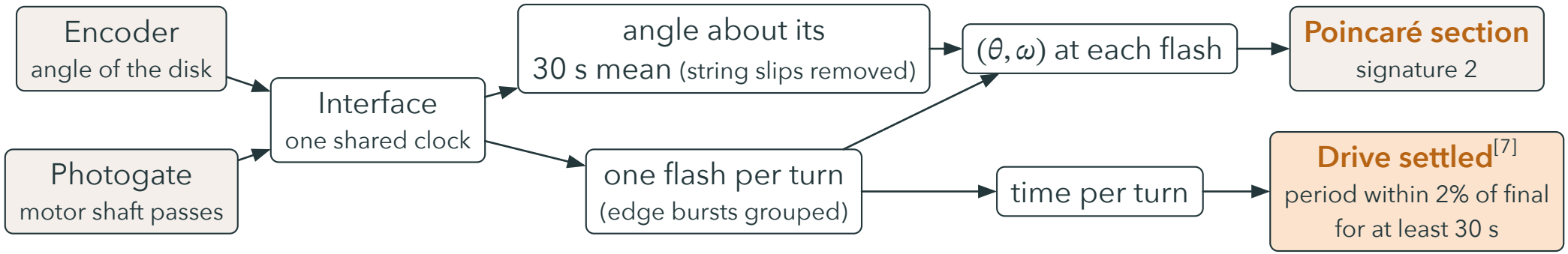


Knob	Sets	In the picture
Motor voltage V	Ω	shaking rate
Drive arm A	γ	shaking strength
Magnet gap h	δ	friction
Added mass m	α, β	valley shape

- **rotary encoder** on the axle: angle θ, speed ω
- **photogate** on the motor shaft: one flash per turn, same clock

Apparatus: PASCO chaotic pendulum, [6] A. Hanks, Chaos, Experiment EX-9907, (n.d.)

From two sensors to strobe pictures and settling times



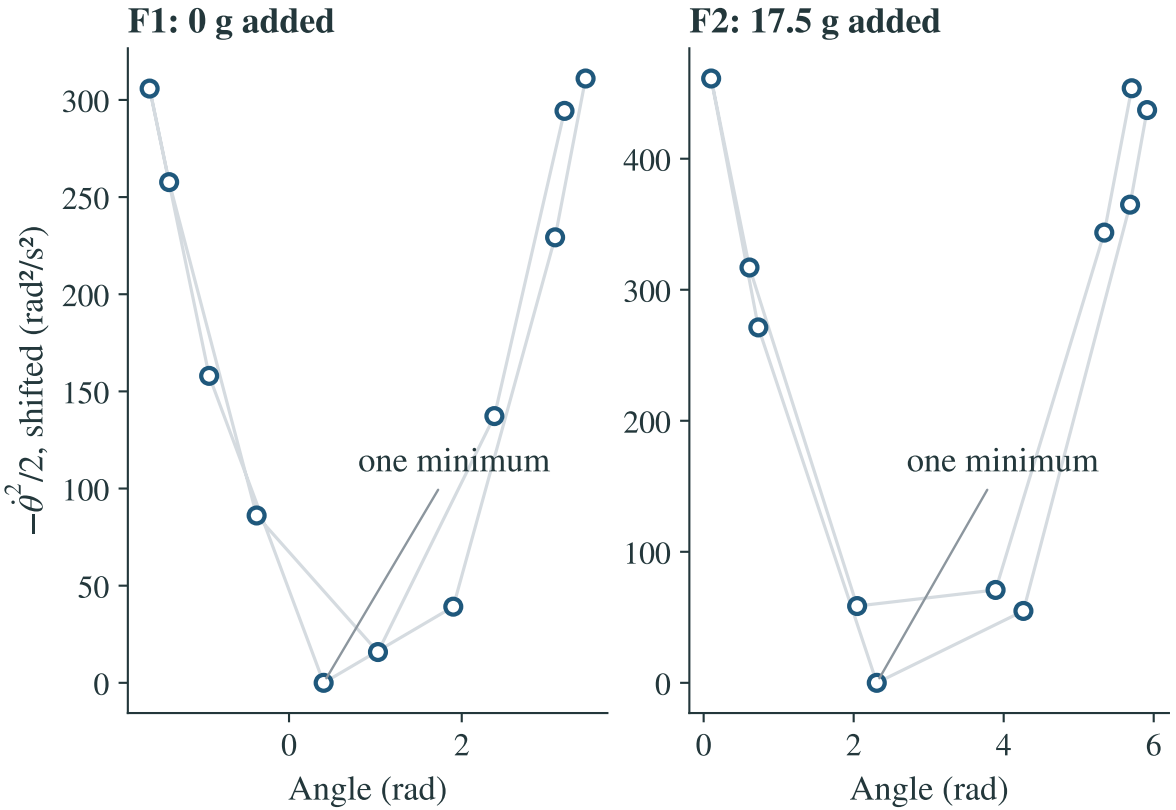
Three kinds of run: motor off, to map the potential · one knob changed at a time · repeated strobed runs

Could the apparatus fake or hide the chaos?

ruled out	encoder steps of 0.004 rad, below any swing	**open**	why the motor speed changes: load or motor itself?
avoided	string slips: none in the compared windows	**limits**	retied strings: “identical” runs differ
robust	2% band: 3% changes no verdict	**assumed**	steady windows (the early one drifts)

[7] W. Irons, Nonlinear Dynamics: The Chaotic Pendulum, PHY 474 laboratory report, The University of Texas at Austin, 2012

One minimum over the sampled angles



Where the pendulum moves faster, the potential is lower.

With little energy loss in one swing,

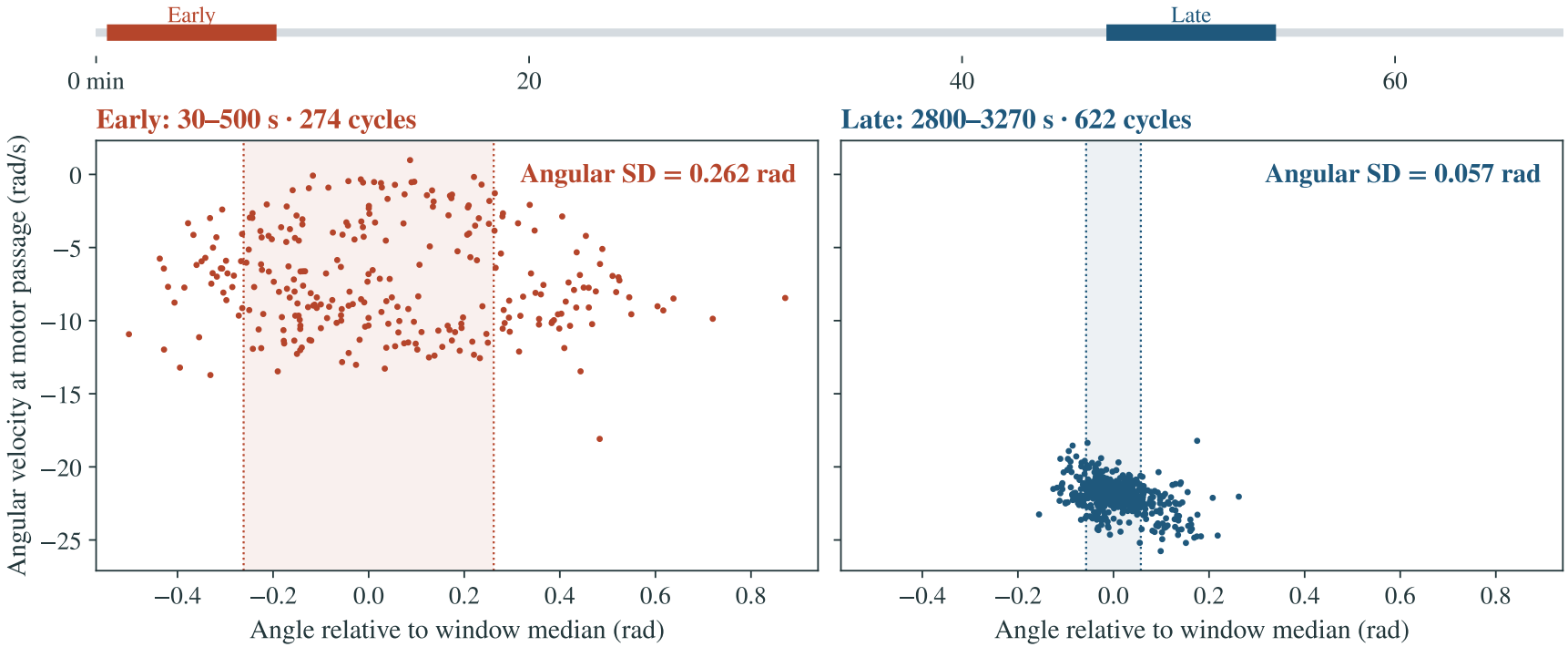
$$\frac{U(\theta)}{I} = \text{constant} - \frac{\dot{\theta}^2}{2}.$$

Both releases resolve one minimum within the angles visited[8].

An unvisited second well is not excluded.

[8] D. J. Suntrup III, Observation of Chaotic Motion in a Nonlinear Mechanical Oscillator, PHY 474 laboratory report, The University of Texas at Austin, 2009
F1: no added mass; F2: 17.5 g added. Released with the motor off, as the procedure for measuring the potential; not written down, but the smooth decay to rest confirms it. Energy loss limits the reconstruction.

The angular spread narrowed by a factor of 4.6



Angular spread: 0.262 → 0.057 rad **4.6× narrower**

Conditional 95% resampling interval: **4.0–5.2**. It does not bound drift or timing errors.

Recording L: two 470 s windows. Spread is angular standard deviation at motor passage; shaded bands show ±1 SD about each window's median. Circular block bootstrap: 16 cycles per block, 4000 resamples; assumes stationarity within each window.

The 0–1 test supports the change in dynamics

Turn the angle samples into a walk.

$$z_{c(n)} = \sum_{j=1}^{n} u_j e^{ijc}$$

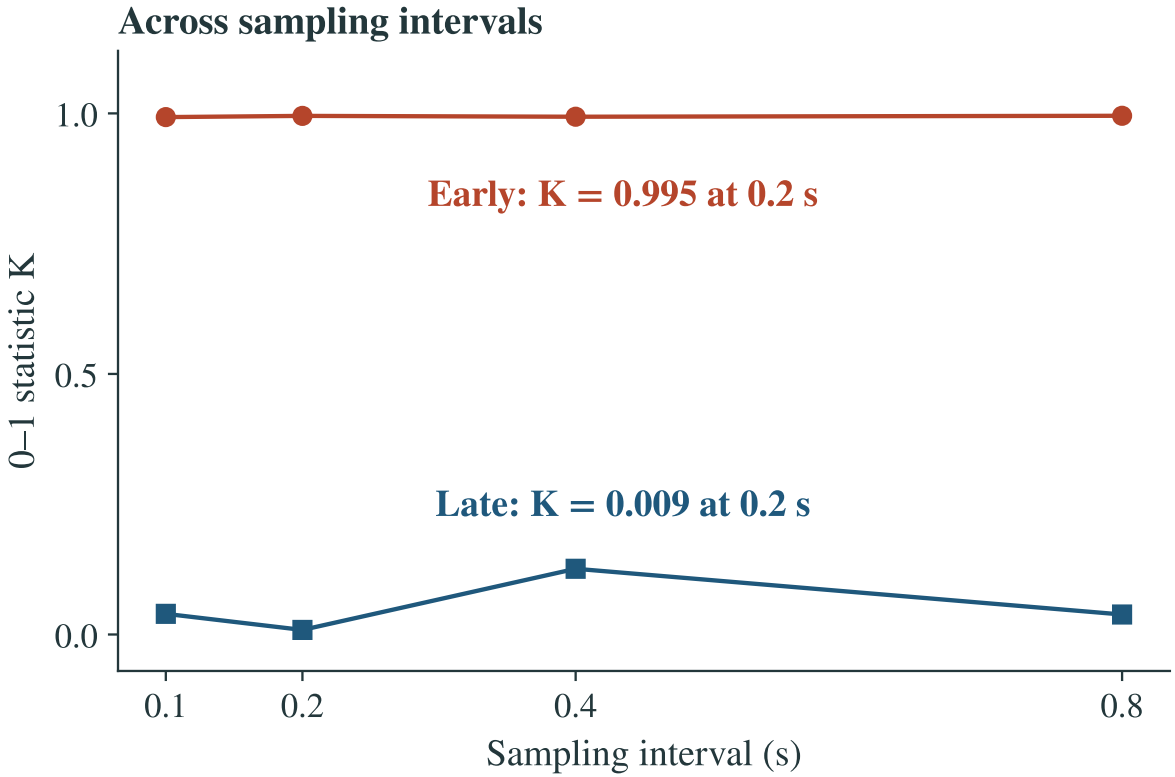
u_j: centered angle sample; c: fixed turn per step.

Does its spread grow with lag?

Spread: mean squared displacement.

$K \approx 0$: bounded spread
$K \approx 1$: roughly linear growth

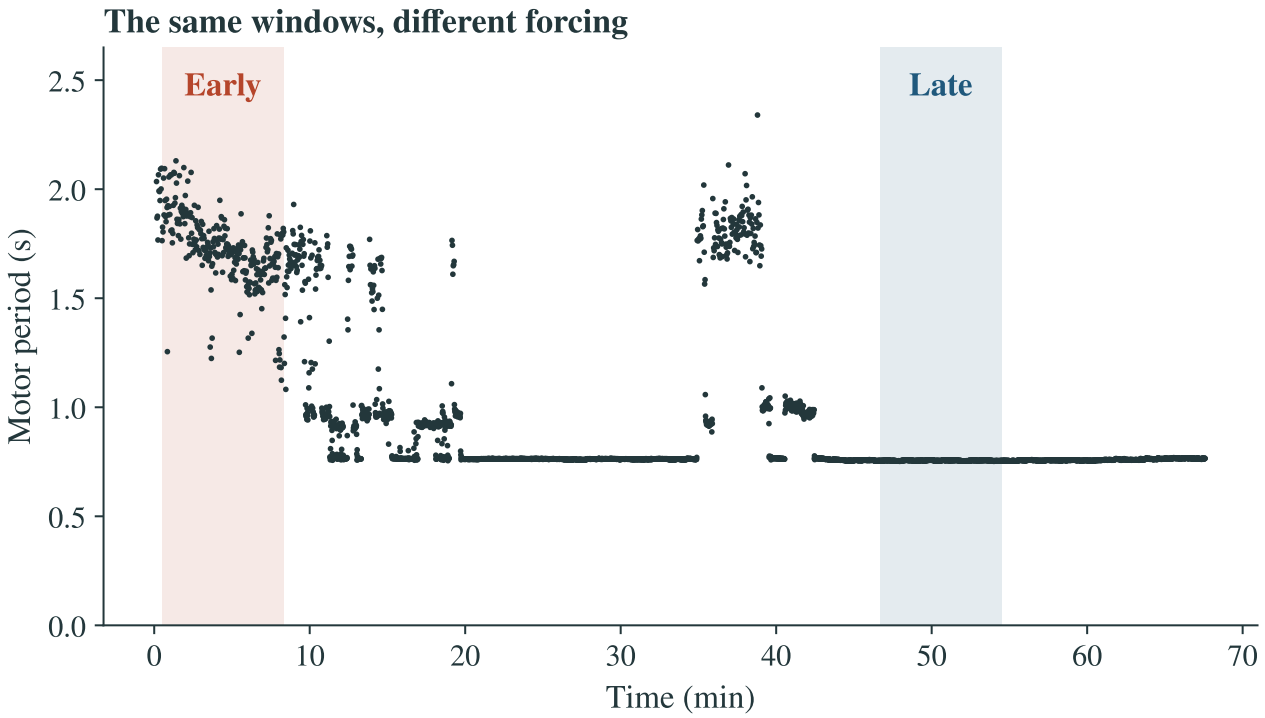
K correlates spread with lag, then takes the median over 100 values of c. It is not a rate or a probability.



White noise also gave $K = 0.998$: a high K alone does not identify chaos.

Uniformly sampled angles from the same 470 s windows; the plotted changes with sampling interval are sensitivity checks, not confidence intervals. Correlation test: [5] G. A. Gottwald and I. Melbourne, On the Implementation of the 0-1 Test for Chaos, SIAM Journal on Applied Dynamical Systems 8, 129 (2009).

The forcing changed despite unchanged settings



Median motor period

1.722 s → 0.756 s

Relative period variation

9.9% → 0.18%

The early and late windows experienced different forcing. The relapse again ran slow and erratic: 0.99 s, 36%.

The narrowing is measured; a transient under **constant forcing** is not established.

Relative variation: standard deviation divided by mean period. Initial 4.98 s startup period is outside the plot range. Even the early window drifts: median period 1.812 → 1.664 s between its halves.

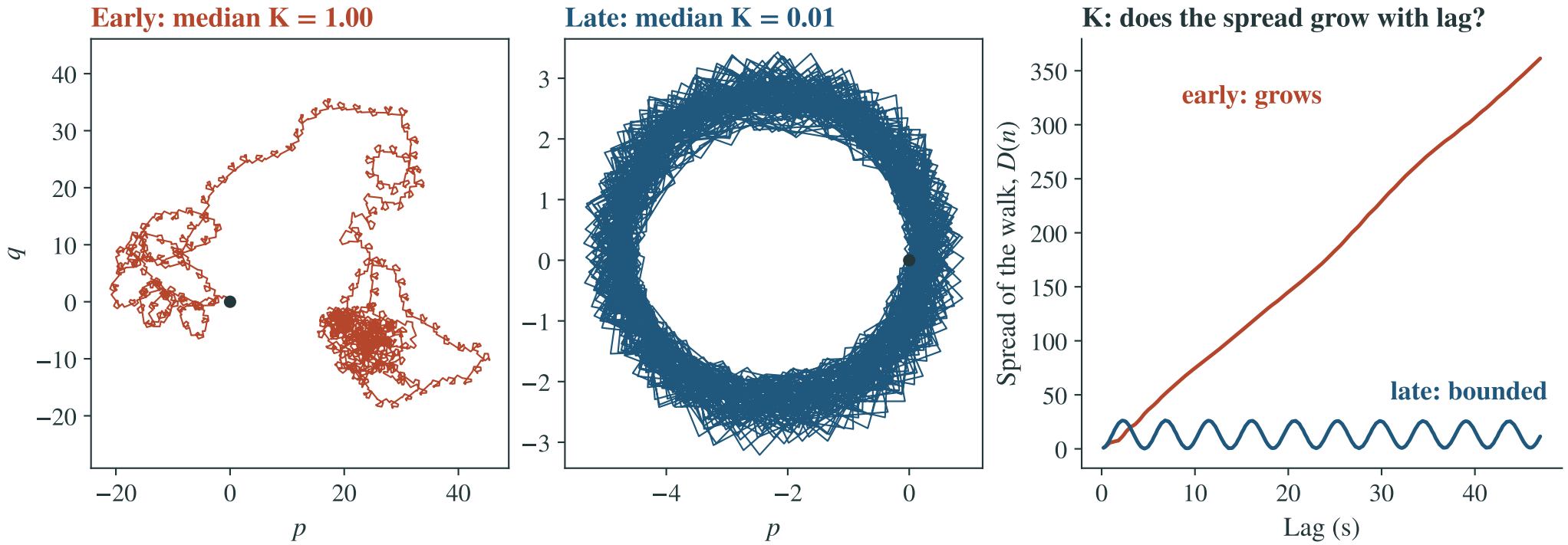
Repeatability increased; its cause remains unresolved



Measured	**4.6× narrower** angular spread Conditional 95% resampling interval: 4.0–5.2
Corroborated	K: **0.995 → 0.009**, supporting the change toward regular motion
Interpretation	Consistent with transient chaos in the coupled apparatus; noise and changing forcing prevent a unique identification.

Next: regulate motor speed; match initial angle, speed and drive phase; measure the noise baseline.

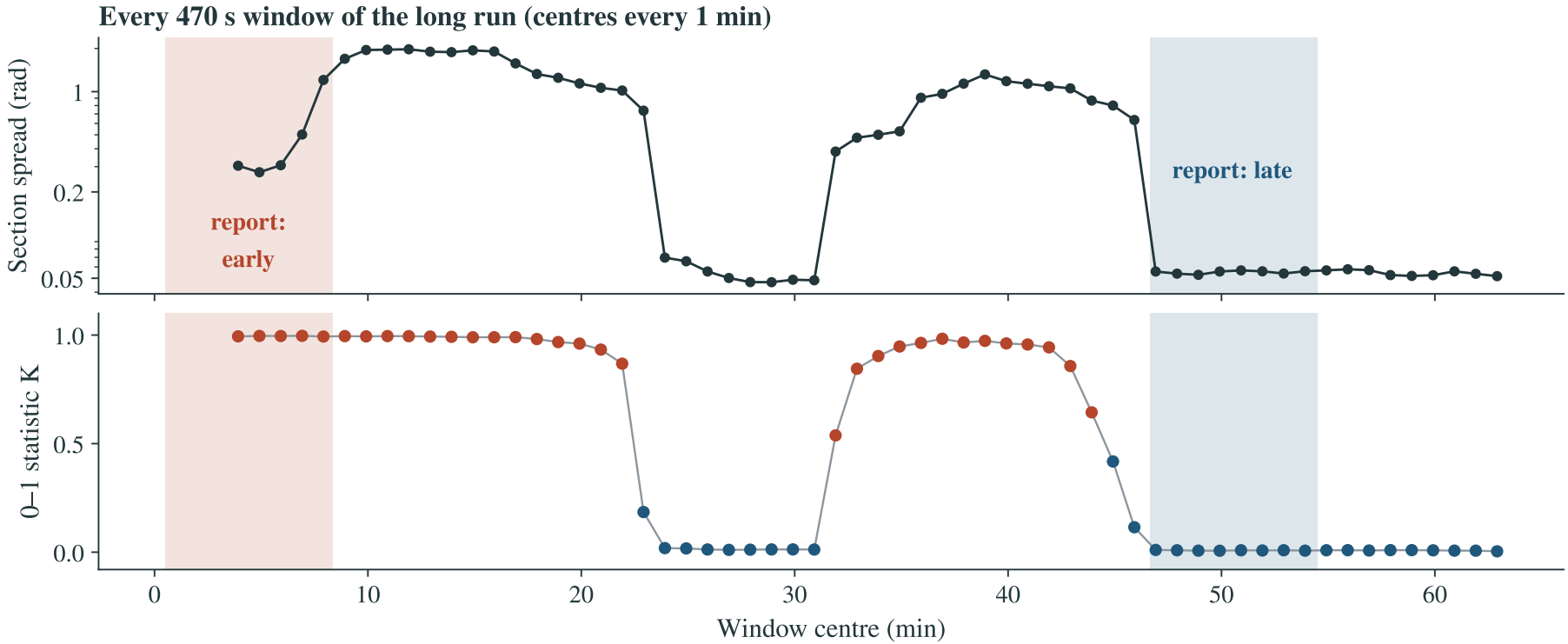
What the 0–1 statistic K measures



Each walk uses one illustrative c. Reported K: median over 100 values of c; lags up to one tenth of the record.

Centered angle samples, scaled by their standard deviation, every 0.2 s. Windows: 30–500 s and 2800–3270 s. Correlation test: [5] G. A. Gottwald and I. Melbourne, On the Implementation of the 0–1 Test for Chaos, SIAM Journal on Applied Dynamical Systems 8, 129 (2009).

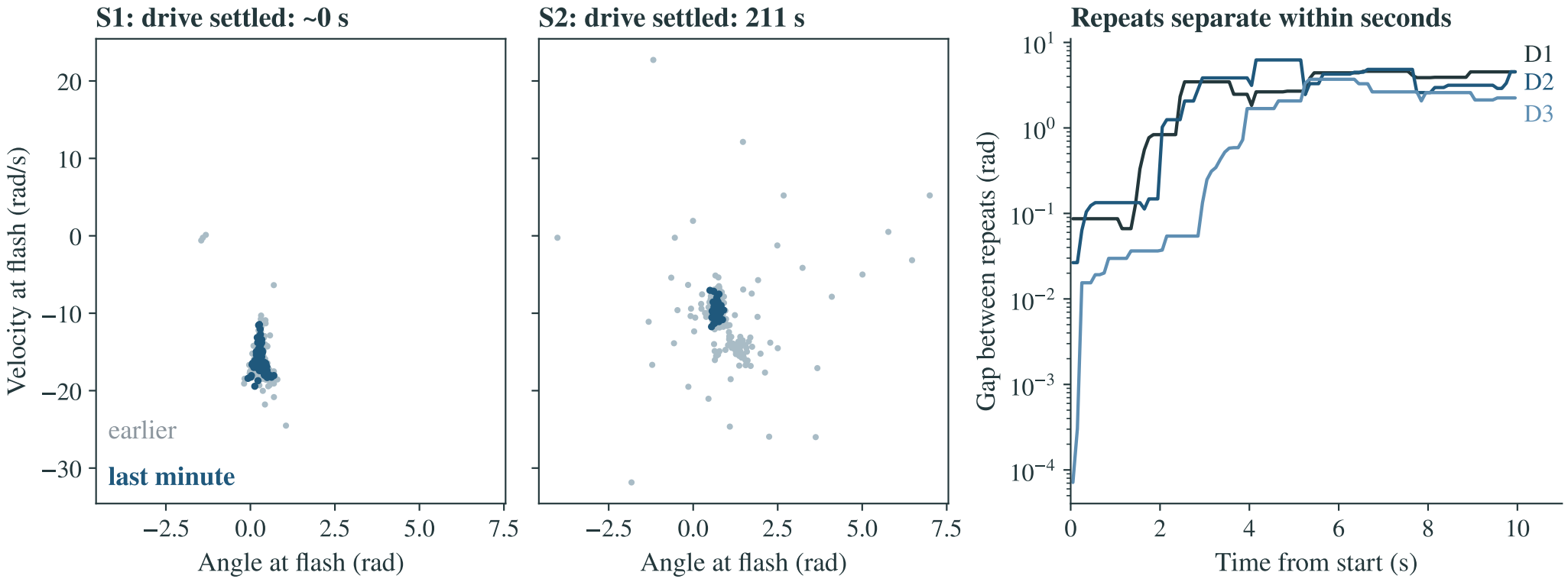
The contrast does not depend on which two windows



Windows inside the first irregular stretch: K 0.99–1.00; the shorter relapse peaks at 0.98. Inside regular stretches: $K \leq 0.02$.

Same functions and 0.2 s sampling as the report; angle about its 30 s running mean because the record slips from about 680 s (inside the report's windows this changes the spread by under 0.002 rad and K not at all). Not part of the report.

Repeated settings did not reproduce the same trajectory



Motor periods met the settling criterion in 29 of 37 records, after 0–2546 s.

Recorded settings matched; initial angle, speed and drive phase did not. Separation curves measure reproducibility. Exploratory exponential rates changed sign under reasonable analysis choices; no Lyapunov exponent is reported.

What the uncertainty and control checks establish

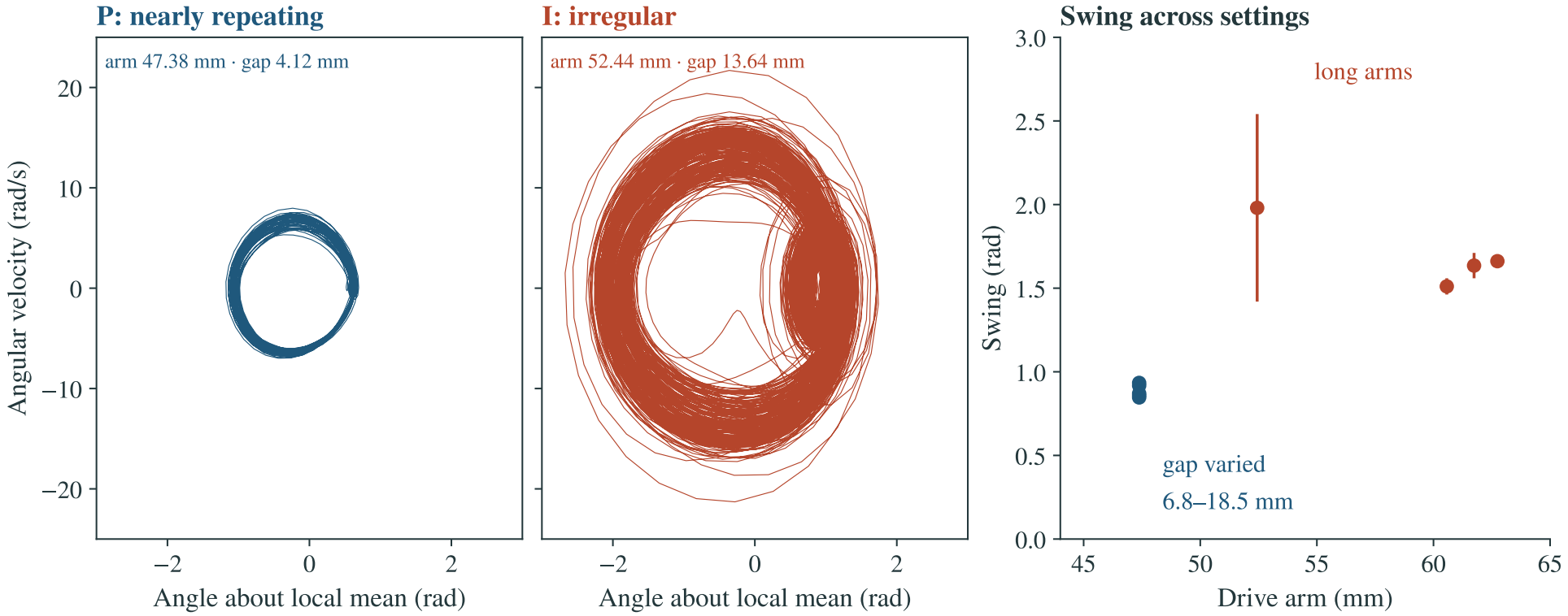
Section-width ratio	4.57; 16-cycle block interval 3.97–5.21. With 8 or 32 cycles, endpoints stayed within 3.87–5.25. Removing a 30 s local mean gave 4.64.
Conditional interval	Finite-record variability assuming stationary windows. Does not cover drive drift, photogate timing errors, or encoder artifacts.
K sensitivity	Sampling 0.1–0.8 s: early 0.993–0.995; late 0.009–0.126. Half-window, lag and phase-increment checks preserve the contrast. These ranges are not confidence intervals.
Failed controls	Once-per-cycle samples gave high K for regular motion. A short periodic reference also approached one at coarse sampling. Use the long, uniformly sampled windows only.

What we can and cannot explain

Observed	Explanation	Status
One minimum	one resolved minimum in each free-release proxy; coefficients not measured	measured
Irregular motion in one valley	a single valley shaken hard can be chaotic[2]	possible
Long-arm series: larger swings	no monotonic arm trend; P/I comparison also changes gap	observed
4.6× narrowing	equal windows; conditional resampling interval 4.0–5.2	measured
K: 1 → 0	supports a dynamics change; noise also gives 1	consistent
Different forcing	median motor period 1.722 vs 0.756 s; cause not isolated	measured
Repeats differ	initial states and string tension not matched	observed
Relapse in the long run	no setting changed; cause not identified	unknown
No period doubling	no systematic bifurcation sweep was performed	not tested

[2] Y. Ueda, Randomly transitional phenomena in the system governed by Duffing's equation, Journal of Statistical Physics 20, 181 (1979)

Changing one setting at a time: long arms, larger swings

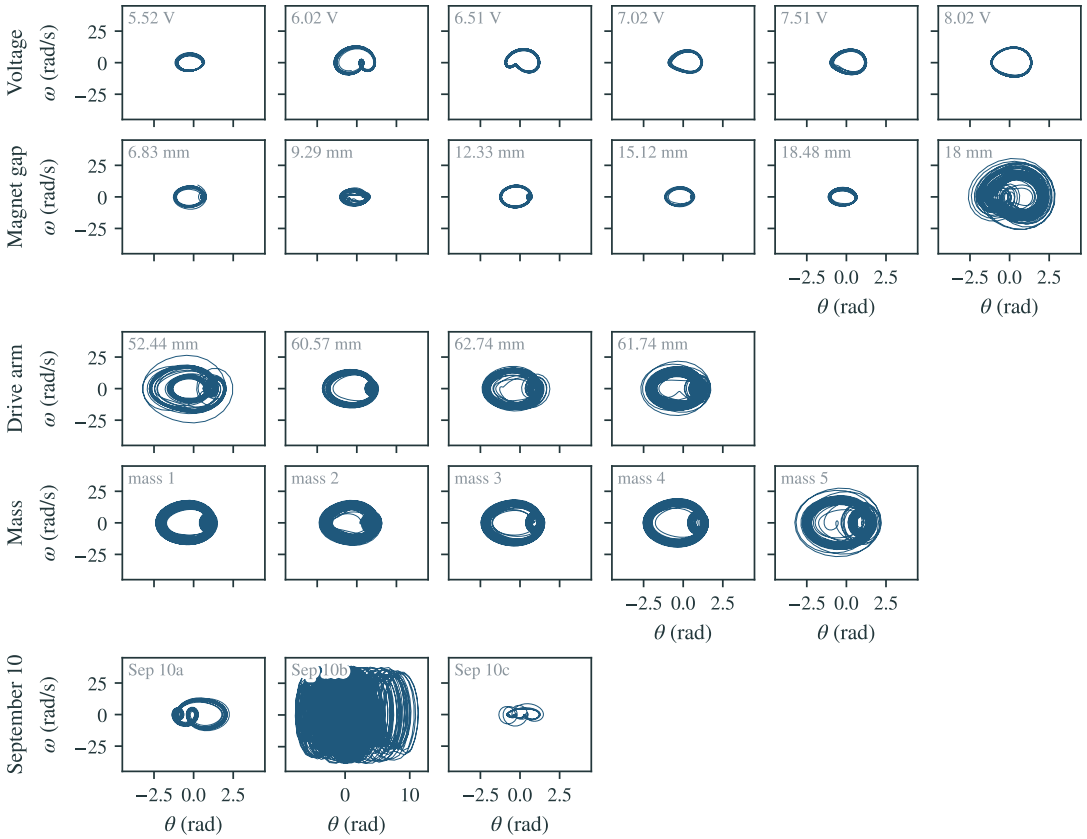


The P/I comparison changes **both arm length and magnet gap**.

Larger swings in the long-arm series; no monotonic increase with arm length.

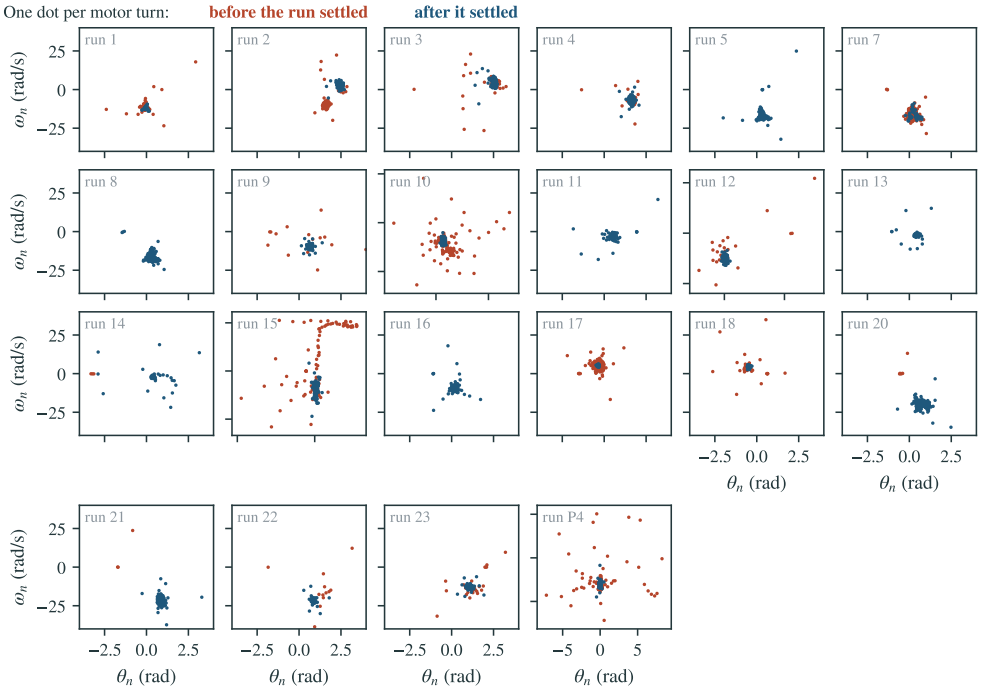
P and I: 5.52 V, 17.5 g added, common portrait scales. Swing: half the 5th–95th percentile range. Bars: half the difference between record halves, not uncertainty between runs.

Every run without the strobe

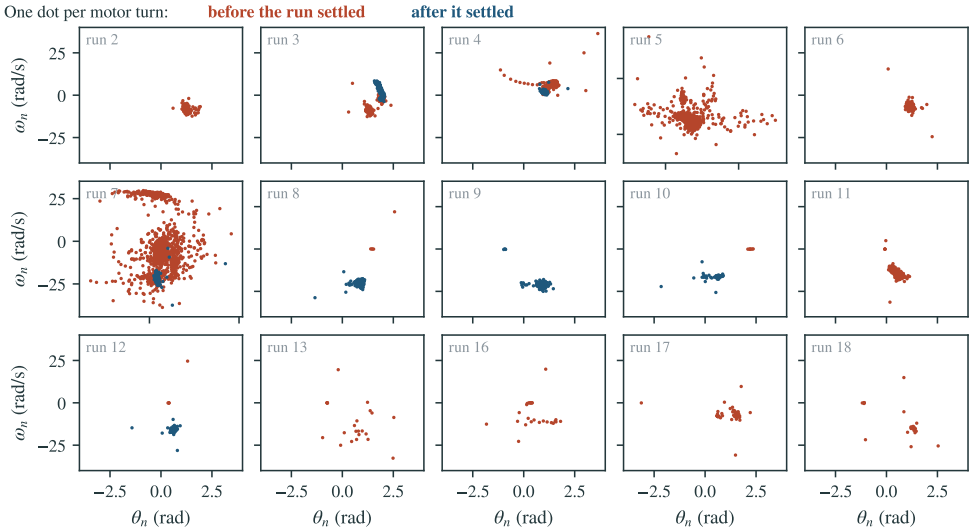


- one loop: periodic
- band of loops: did not repeat
- repeated settings can give a single loop in one run and a filled plane in the next

Strobe pictures of every photogate run

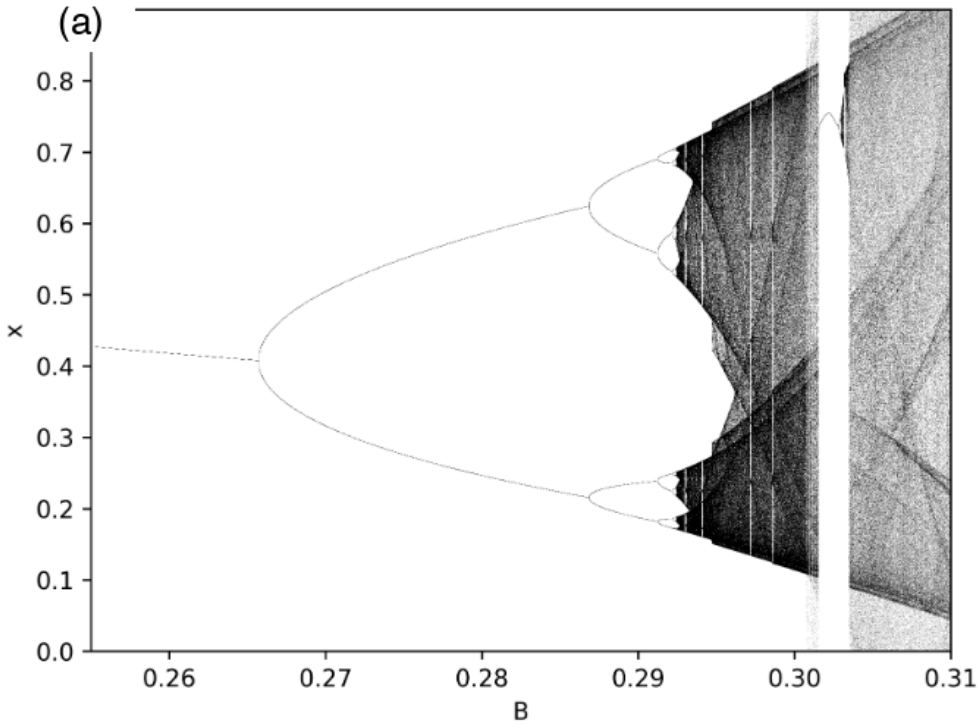


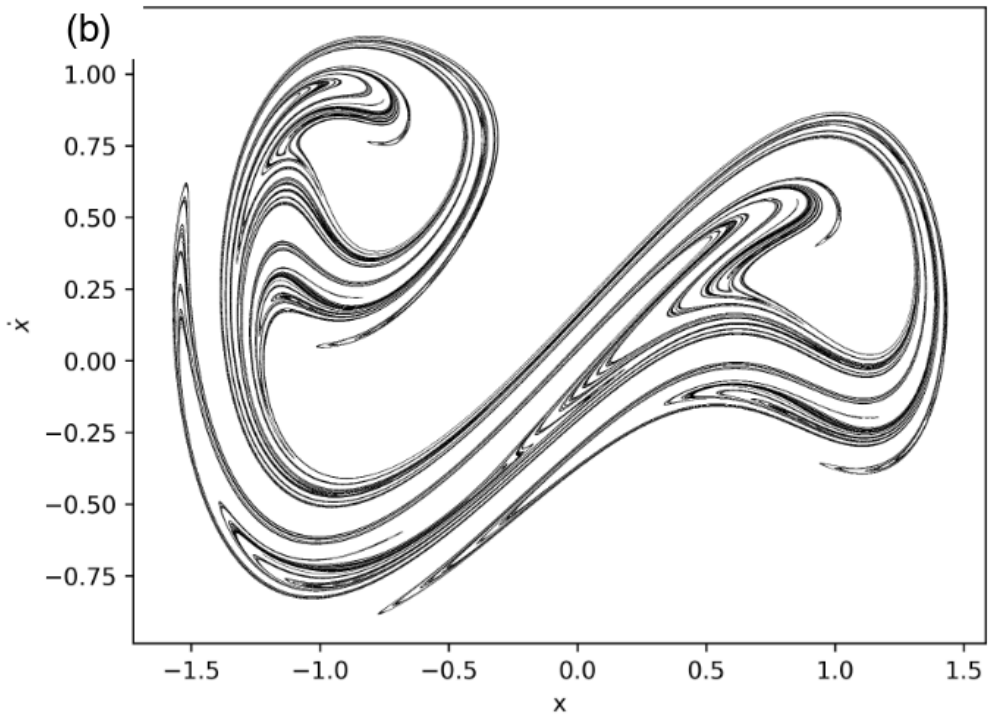
Day 1: every run ends in one tight dot, after 0 to 479 s as a cloud.



Day 2: some dots form while the motor still wanders; one run stays a cloud for 14 min; the shortest are too short to judge.

What period doubling would have looked like





Simulation of a two-valley Duffing oscillator[9], not our data. (a) One dot splits into two, four, then a band. (b) The folded chaotic cloud.

[9] J. Jani, Numerical Simulation and Analysis of the Duffing System using Python, WSEAS Transactions on Systems 23, 301 (2024)

References

[1] F. C. Moon and P. J. Holmes, A magnetoelastic strange attractor, Journal of Sound and Vibration **65**, 275 (1979).

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[6] A. Hanks, Chaos, Experiment EX-9907, (n.d.).

[7] W. Irons, Nonlinear Dynamics: The Chaotic Pendulum, PHY 474 laboratory report, The University of Texas at Austin, 2012.

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